Activity 14 Reflection and Evidence

Blake Cantave | CSC 4360 | November 9, 2026

Repository: Prophet-20/In-Class-Activity-14

Setup and payload

I kept Firebase Messaging on version 16.x to match the assignment and used the Android app registered with my Firebase project. The two data keys make the screen change straightforward: action must be update_ui, and asset selects a bundled image. With asset set to promo, the app shows the delivery van and "Out for delivery." Keeping a fixed list of supported assets prevents a message from loading an arbitrary file.

Lifecycle and evidence

The screenshots made the different message paths easier to follow. T1 used onMessage while the app was visible. T2 used onMessageOpenedApp after the notification was tapped from the background. T3 started a new app process and used getInitialMessage, with the receipt count resetting to 1. The T1-T3 evidence shows each handler and its result. Background processing runs separately from the widget tree on Android, so that callback cannot call setState to update the screen directly.

Validation and next step

For T4, changing only asset to unknown produced a message explaining that the asset was unsupported. The previous title and van image stayed visible, and the app kept running. One earlier issue was an extra lib folder that broke the service import; moving fcm_service.dart beside main.dart fixed it. My next improvement would be fetching the current order status from a backend after a notification, so the displayed information stays authoritative.

Test details and evidence

All four required tests passed on the Pixel 10 Pro XL Android emulator. The four labeled evidence images are linked below and included in the repository.

Device and permission

Android 17 / API 37; emulator-5554; reported model sdk_gphone16k_arm64. Notification permission: authorized; POST_NOTIFICATIONS granted. Test session: November 9, 2026, 7:11-7:15 PM EST.

Valid payload

T1-T3 reused this notification and custom string data. This illustrates the message content, not a complete HTTP v1 request.

```
{
  "notification": {
    "title": "Order #1001 is out for delivery",
    "body": "Your driver is on the way."
  },
  "data": {
    "action": "update_ui",
    "asset": "promo"
  }
}
```

T4 changed only data.asset from promo to unknown.

Four result rows

Test	Expected	Observed	Status	Evidence
T1	onMessage and a visible update	Count 2; delivery title and van image appeared.	PASS	T1-evidence.png
T2	onMessageOpenedApp after a background tap	Count 3; notification tap returned to the existing process.	PASS	T2-evidence.png
T3	getInitialMessage on a notification cold start	New process; count 1; delivery UI loaded.	PASS	T3-evidence.png
T4	onMessage with rejection and preserved content	Count 2; unsupported-asset diagnostic; previous title and image retained; no crash.	PASS	T4-evidence.png

Counts progressed 2, 3, then reset to 1 at T3 and increased to 2 at T4. One earlier receipt existed before T1.

Limitations

These results cover one Android emulator. iOS and other platforms were not tested. Network conditions and device policies can affect delivery. No app code or configuration changes were needed during the four tests.

Evidence packet: View the screenshots and summary